

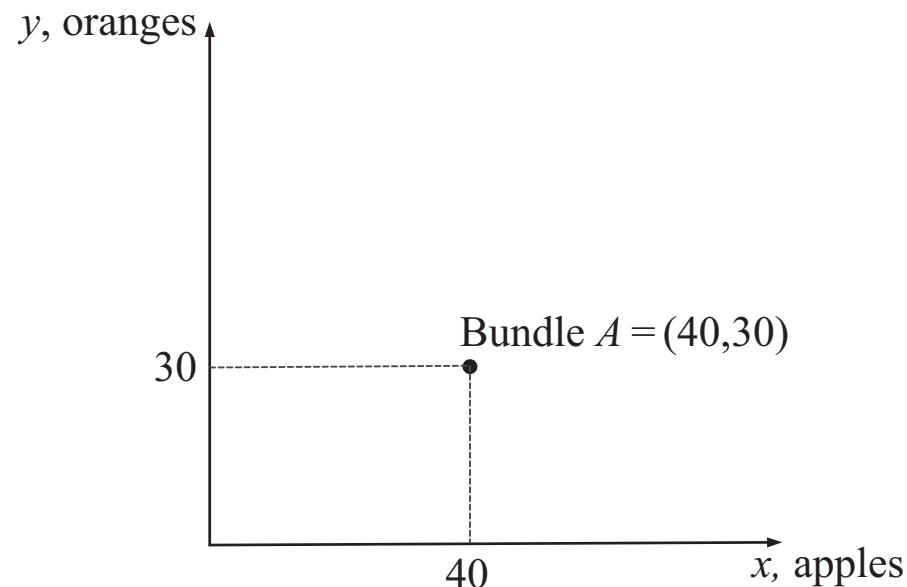
ECO211: Microeconomics I

Consumer Preferences and Utility

Bundles

Bundles

- A **bundle** is a list of goods and services.
 - *Example:* If an individual consumes only 2 goods, x and y (apples and oranges), bundle $A = (40, 30)$ indicates that she consumes $x = 40$ apples and $y = 30$ oranges.



Preferences for Bundles

- Let's analyze consumer preferences over bundles or **how a consumer ranks different bundles**.
- Notation for comparing preferences for bundles:
 - Consider bundles $A = (x_A, y_A)$ and $B = (x_B, y_B)$.
 - $A \succ B$, the individual “strictly prefers” bundle A to B (“strictly” rules out the possibility she is indifferent between the two bundles).
 - $A \sim B$, she “indifferent” between bundles A and B .
 - $A \succeq B$, she “weakly prefers” bundle A to B (she can be indifferent between the two bundles or strictly prefer A to B).

Preferences for Bundles

- Completeness:

- A preference relation is *complete* if the consumer has the ability to compare any two bundles A and B :
 - $A \succ B$ (she strictly prefers bundle A),
 - $B \succ A$ (she strictly prefers bundle B), or
 - $A \sim B$ (she is indifferent between A and B).
- We don't allow the consumer to respond "I don't know how to compare these two bundles!"

Preferences for Bundles

- Transitivity:

- For every three bundles A , B , and C ,
 - if the consumer prefers A to B ($A \succ B$),
 - and B to C ($B \succ C$),
 - she must also prefer A to C ($A \succ C$).
- A consumer with intransitive preferences would have $A \succ B$ and $B \succ C$, but $C \succ A$. Her preferences would exhibit a cycle:

$$A \succ B \succ C \succ A.$$

- An individual with intransitive preferences would be subject to exploitation.

Preferences for Bundles

- Exploitation of intransitive individuals:

- Consider 3 goods, an orange, and apple, and a banana. And a consumer with the following preferences:

Orange $\succ$ *Banana* and *Banana* $\succ$ *Apple*

but *Apple* $\succ$ *Orange* (which violates transitivity)

- Assume she owns 1 orange, and she plays a game with a fruit seller. If the seller gives her preferred fruit, she pays Rs. 10.
 - The seller offers an apple, she pays Rs. 10 because *Apple* $\succ$ *Orange*.
 - Next, the seller offers a banana, she pays Rs. 10 because *Banana* $\succ$ *Apple*.
 - Next, the seller offers an orange, she pays Rs. 10 given *Orange* $\succ$ *Banana*.
- At the end, the consumer has 1 orange as at the beginning of the exchange, but she has lost Rs. 30 due to her intransitive preferences.

Preferences for Bundles

- Strict Monotonicity:
 - Consider an initial bundle A , and a new bundle B , where bundle B has
 - the same amount of good x as bundle A ($x_A = x_B$),
 - but more units of good y ($y_B > y_A$).
 - A consumer's preferences satisfy *strict monotonicity* if
 - she strictly prefers B to A ($B \succ A$).
 - Increasing the units of a single good, as y in bundle B , produces a new bundle that is strictly preferred to $A \rightarrow$ “more of *anything* is strictly preferred.”

Preferences for Bundles

- **Monotonicity:**
 - Again, consider an initial bundle A , and a new bundle B , where bundle B has the same amount of good x as bundle A ($x_A = x_B$), but more units of good y ($y_B > y_A$),
 - whereas a new bundle C has more units of both goods than bundle A does ($x_C > x_A$ and $y_C > y_A$).
 - A consumer's preferences satisfy *monotonicity* if
 - she weakly prefers B to A ($B \succeq A$),
 - but she strictly prefers C to A ($C \succ A$)
 - If the amounts of *all* goods are higher, as in bundle C , the consumer is better off → “more of *everything* is strictly preferred.”

Preferences for Bundles

- *Example: Monotonic and strictly monotonic preferences.*
 - Consider bundles $A = (2,3)$ and $B = (2,4)$.
 - Gopal strictly prefers bundle B to A ($B \succ A$).
 - If this ranking holds for any two bundles where only one of the good is increased, these preferences satisfy *strict monotonicity*.
 - Vinayak is indifferent between B and A ($B \sim A$).

If we replace B with bundle $C = (3,4)$, he strictly prefers C to A ($C \succ A$).

 - If this ranking holds for any two bundles in which one has more units of all goods, his preferences satisfy *monotonicity*.

Preferences for Bundles

- Strict monotonicity implies monotonicity, but not the reverse
- Monotonicity and strict monotonicity require that the consumer regards all items in her bundle as “goods” rather than “bads” (e.g., pollution or garbage).
 - If some good were a bad, increasing the number of units in initial bundle A , would produce a new bundle B that would be *less* preferred than bundle A , violating monotonicity and strict monotonicity.

Preferences for Bundles

- Nonsatiation:

- Preferences satisfy nonsatiation if, for every bundle A , there is another bundle B for which the consumer is better off,

$$B \succ A$$

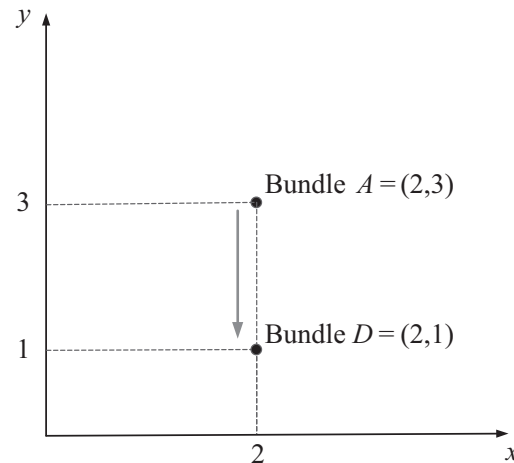
- Nonsatiation means there is no “bliss bundle”.
- Nonsatiation allows the consumer to regard some goods as “bads.”
- Nonsatiation only requires the consumer to always find more preferred bundles.

Monotonicity $\Rightarrow$ Nonsatiation

Monotonicity $\nLeftarrow$ Nonsatiation

Preferences for Bundles

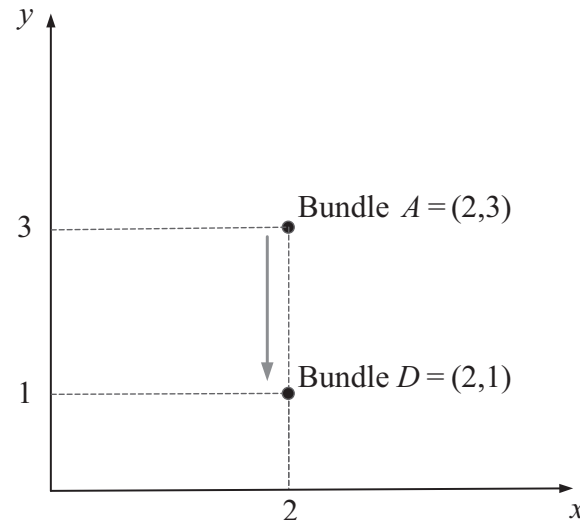
- *Example: Nonsatiated preferences.*
 - Consider bundles $A = (2,3)$ and $D = (2,1)$.



- Gopal says he strictly prefers D to A , $D \succ A$
Do his preferences satisfy nonsatiation?

Preferences for Bundles

- *Example* (continued):
 - His preferences, $D \succ A$, can satisfy nonsatiation, but violate monotonicity.
 - Relative to bundle A , bundle D decreased the amount of good y , keeping x unaffected.
 - If $D \succ A$, it must be that y is a bad.
 - Bundle D is a “more preferred” bundle $\rightarrow$ nonsatiation is not violated.
 - Monotonicity would require $A \succ D$.



Utility Functions

Utility Functions

- A **utility function** mathematically represents a preference relation
- *Example:* If a consumer consumes bundle $A = (40,30)$ and her utility function is $u(x, y) = 3x + 5y$, her level of utility at bundle A is

$$u(40,30) = (3 \times 40) + (5 \times 30) = 270$$

- The utility level from bundle A is not important. Only the *ranking* of utilities is important:
 - Only the utility ranking matters $\rightarrow$ “ordinality.”
 - The specific utility level that the consumer reaches with each bundle matters $\rightarrow$ “cardinality.”

Utility Functions

- *Example: Utility ranking and increasing transformations of the utility function.*
 - Consider utility function $u(x, y) = xy$:
 - Bundle $A = (40, 30)$ produces $u(40, 30) = 1,200$.
 - Bundle $B = (20, 30)$ generates $u(20, 30) = 600$.
 - The consumer prefers bundle A to B ($A \succsim B$).
 - Consider now utility function $v(x, y) = 3xy + 8$, which is an increasing transformation of $u(x, y)$:
 - Bundle $A = (40, 30)$ yields $v(40, 30) = 3,608$.
 - Bundle $B = (20, 30)$ still generates $v(20, 30) = 1,808$.
 - The consumer stills prefers bundle A to B ($A \succsim B$).
 - The consumer's preference over bundle A and B is unaffected (i.e., her ranking does not change). Thus, both utility functions represent the same preference relation.

Utility Functions

- *Example: Testing properties of preference relations.*

Consider utility function $u(x, y) = xy$. We check:

a) Completeness:

- For every two bundles, $A = (x_A, y_A)$ and $B = (x_B, y_B)$, completeness holds when
 - either $u(x_A, y_A) > u(x_B, y_B)$,
 - $u(x_B, y_B) > u(x_A, y_A)$, or
 - $u(x_A, y_A) = u(x_B, y_B)$.
- This is true for $u(x, y) = xy$

Utility Functions

- *Example* (continued):

b) Transitivity:

- For every three bundles, A , B , and C , where $u(x_A, y_A) \geq u(x_B, y_B)$ and $u(x_B, y_B) \geq u(x_C, y_C)$, transitivity holds when
 - $u(x_A, y_A) \geq u(x_C, y_C)$
- This is true for $u(x, y) = xy$

Utility Functions

- *Example* (continued):

- c) *Strict monotonicity:*

- Consumers with strictly monotonic preferences prefer bundles with more units of *any* good.
 - For this property to hold, we need $u(x, y) = xy$ to be *strictly* increasing in both goods. Is this true?

$$\frac{\partial u(x,y)}{\partial x} = y \geq 0 \text{ and } \frac{\partial u(x,y)}{\partial y} = x \geq 0$$

- Increasing the units of x produces a strict increase in consumer's utility as far as $y > 0$.
 - If she does not consume good y at all, $y = 0$, increasing good x does not alter utility level.
 - Therefore, strict monotonicity does not hold because an increase in x does not necessarily increase consumer's utility.

Utility Functions

- *Example* (continued):

- d) *Monotonicity*:

- We need $u(x, y) = xy$ to be *weakly* increasing in x and y .
 - We know that an increase in x
 - produces a strict increase in consumer's utility (when $y > 0$),
 - or does not affect utility (when $y = 0$),
 - but it never reduces utility.

Utility Functions

- *Example* (continued):

- d) *Monotonicity* (cont.):

- An increase in both x and y produces a new bundle that generates a strictly greater utility.
 - Consider good x is increased by $a > 0$ and good y by $b > 0$. This yields a utility level of

$$u(x + a, y + b) = (x + a)(y + b).$$

- Monotonicity is satisfied because

$$u(x + a, y + b) > u(x, y).$$

Utility Functions

- *Example* (continued):

e) Nonsatiation:

- This property holds by monotonicity.
- We found that increasing amounts of both goods produces a new bundle $(x + a, y + b)$, that is strictly preferred to the original bundle (x, y) .
- Starting from the original bundle we can always find another bundle for which the consumer is better off.
- The consumer is never satiated.

Utility Functions

- *Utility functions and their properties.*

Utility Function	Completeness	Transitivity	Strict Monotonicity	Monotonicity	Nonsatiation
$u(x, y) = by$					
$u(x, y) = ax$					
$u(x, y) = ax - by$					
$u(x, y) = ax + by$					
$u(x, y) = A \min\{ax, by\}$					
$u(x, y) = Ax^\alpha y^\beta$					

Parameters a, b, A, α, β are positive.

Marginal Utility

Marginal Utility

- **Marginal utility of a good** is the rate at which utility changes as the consumption of a good increases.
 - Intuitively, *how much better off do you become by consuming 1 more unit of good x ?*
 - Mathematically, marginal utility of good x is

$$MU_x = \frac{\partial u(x, y)}{\partial x},$$

and similarly for good y , $MU_y = \frac{\partial u(x, y)}{\partial y}$.

- Graphically, we measure the slope (rate of change) of the utility function as we increase the amount of good x , holding the amount of other goods constant.

Marginal Utility

- *Example: Finding marginal utility, MU.*

- Consider utility function $u(x, y) = x^{1/2}y^{1/2}$.
- Marginal utility of good x is

$$MU_x = \frac{1}{2}x^{\frac{1}{2}-1}y^{\frac{1}{2}} = \frac{1}{2}x^{-\frac{1}{2}}y^{\frac{1}{2}}$$

- Rearranging,

$$MU_x = \frac{1}{2} \frac{y^{1/2}}{x^{1/2}}.$$

- $MU_x > 0$ when the individual consumes positive amounts of good x and y , indicating that 1 more unit of good x raises her utility.

Marginal Utility

- Diminishing Marginal Utility.

- Marginal utilities of most utility functions are *decreasing* in the amount of the good that the individual consumes,

$$MU_x \text{ decreases in } x, \text{ or } \frac{\partial MU_x}{\partial x} \leq 0 \text{ (similarly for } y\text{).}$$

- While more units of good x increase utility level, further increments in x produce smaller utility gains.
 - When the consumer has few units of good (e.g., food), giving her with 1 more unit increases her utility a great deal.
 - When she already has large amounts, giving her 1 more unit of food produces a small utility gain (or no gain at all!)

Marginal Utility

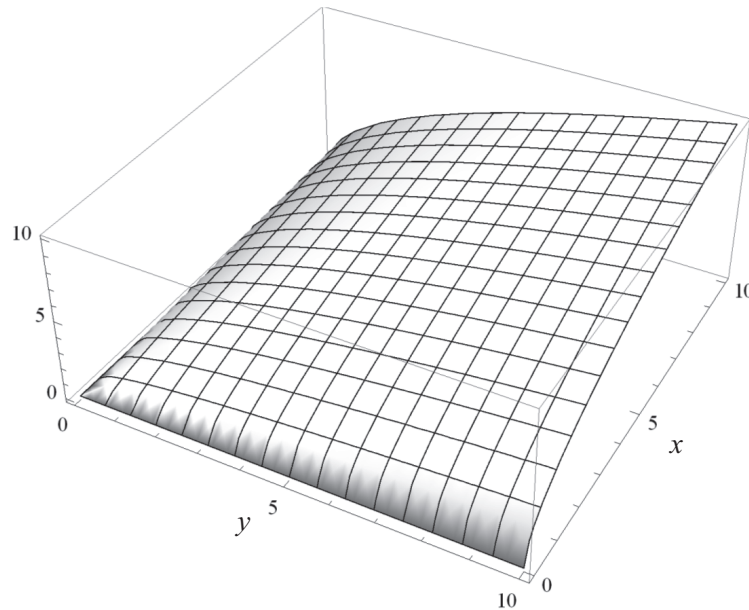
- *Example: Diminishing marginal utility.*
 - Consider $u(x, y) = x^{1/2}y^{1/2}$.
 - Marginal utility of good x was $MU_x = \frac{1}{2} \frac{y^{1/2}}{x^{1/2}}$.
 - MU_x is decreasing in the amount the consumer enjoys of good x ,

$$\frac{\partial MU_x}{\partial x} = -\frac{y^{1/2}}{4x^{3/2}} < 0 \text{ for all values of } x \text{ of } y.$$

Indifference Curves

Indifference Curves

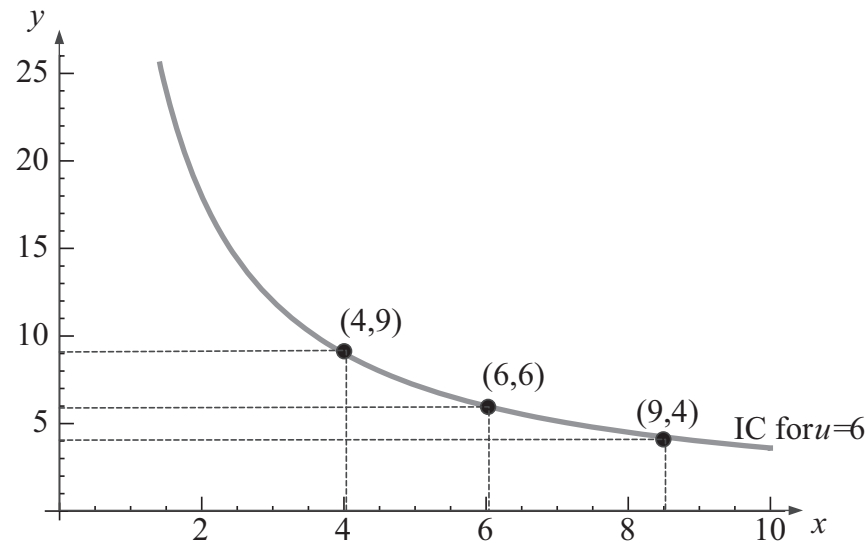
- This figure depicts $u(x, y) = x^{1/2}y^{1/2}$.
 - The height of the “mountain” is the utility that the individual achieves by consuming a specific amount of x and y .



- At bundle $(x, y) = (4, 9)$,
 $u(4, 9) = 4^{1/2}9^{1/2} = 6$.
- This utility level can also be obtained at bundles:
 - $(x, y) = (6, 6)$,
 $u(6, 6) = 6^{1/2}6^{1/2} = 6$.
 - $(x, y) = (9, 4)$,
 $u(9, 4) = 9^{1/2}4^{1/2} = 6$.

Indifference Curves

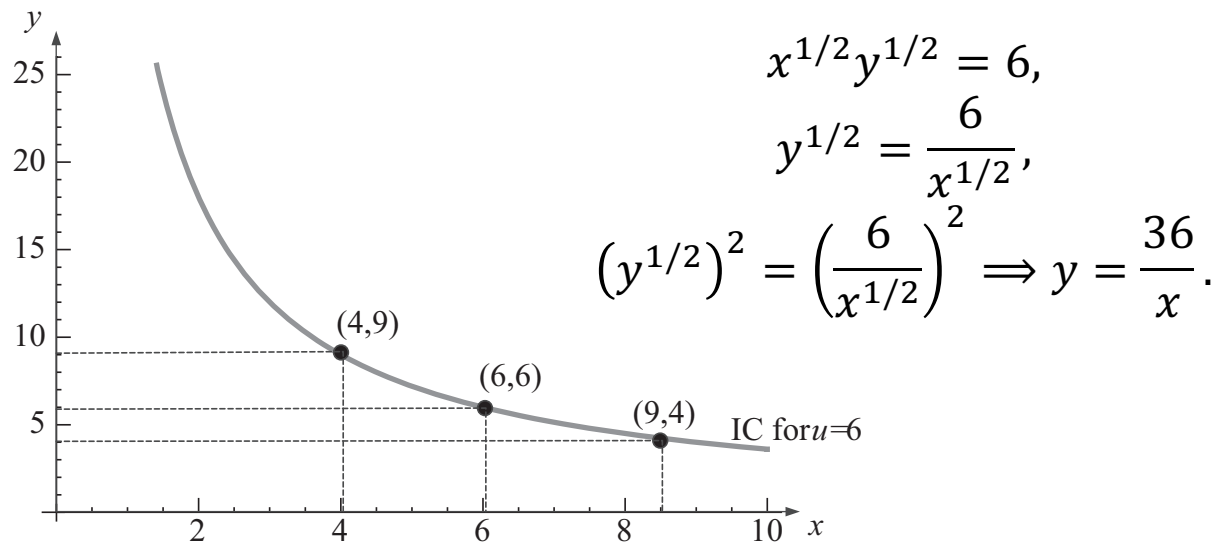
- The next figure depicts a “slice” of the utility mountain at a height of $u = 6$.



- This curve connects bundles at which the consumer obtains the same utility $u = 6$. She is indifferent between consuming any of these bundles.

Indifference Curves

- **Indifference curve (IC):** A curve connecting consumption bundles that yield the same utility level.
 - IC for $u(x, y) = x^{1/2}y^{1/2}$ evaluated at $u = 6$. Solving for y , we find the expression of the indifference curve:



Indifference Curves

- Consider again utility function $u(x, y) = x^{1/2}y^{1/2}$.
- We want to obtain the expression for the indifference curve when the consumer reaches utility level $u = 10$.

- This indifference curve entails

$$x^{1/2}y^{1/2} = 10.$$

- Solving for y ,

$$y^{1/2} = \frac{10}{x^{1/2}}.$$

- Squaring both sides we obtain the indifference curve:

$$(y^{1/2})^2 = \left(\frac{10}{x^{1/2}}\right)^2 \Rightarrow y = \frac{100}{x}.$$

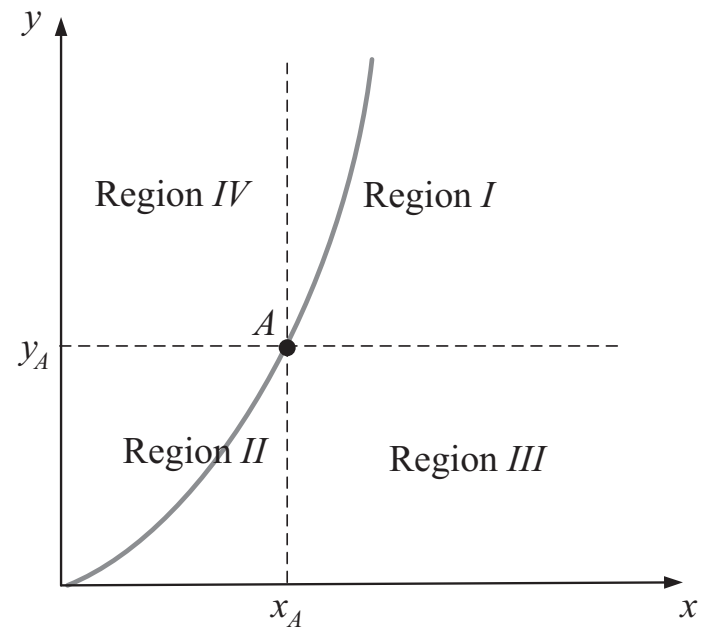
Indifference Curves

- *Example* (continued):
 - Consider now $u(x, y) = 5x + 3y$, and $u = 9$.
 - Solving for y in $5x + 3y = 9$,

$$y = 3 - \frac{5}{3}x.$$

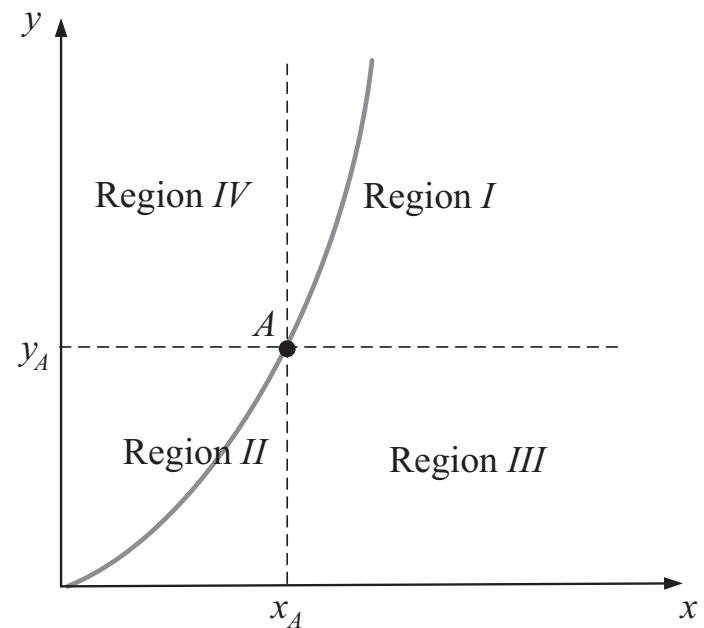
Properties of Indifference Curves

- ICs are not positively sloped. Consequence of monotonicity.
 - Consider bundle $A = (x_A, y_A)$ in a positively sloped IC.
 - The IC passing through bundle A cannot go through Regions I and II because the consumer strictly prefers
 - bundles in Region I than A ,
 - bundle A than those in Region II.



Properties of Indifference Curves

- ICs are not positively sloped.
- The IC passing through bundle A can only go through Region III and IV.

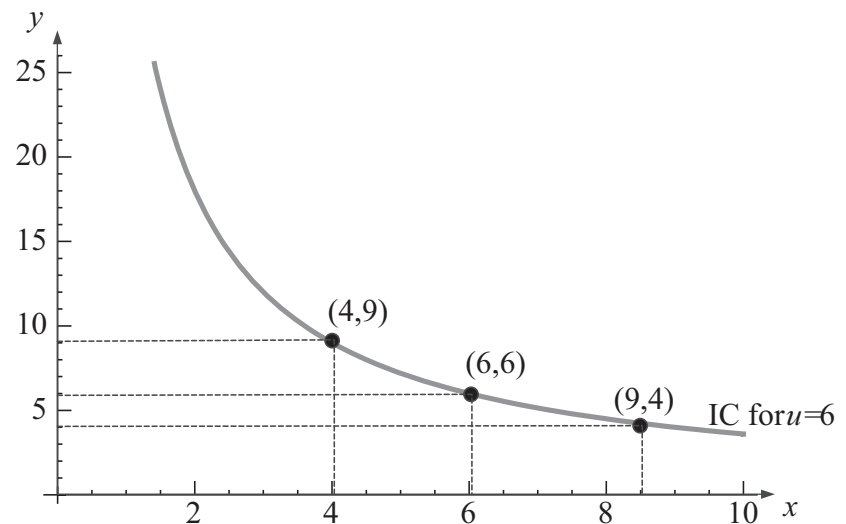


Properties of Indifference Curves

- ICs are convex.

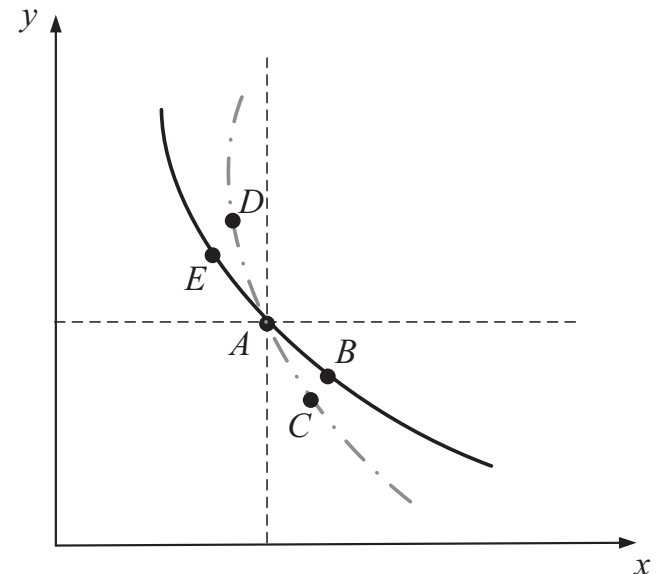
For any two bundles, a straight line connecting them lies:

- (1) strictly above the curve, yielding a higher utility (when the IC curve is strictly decreasing); or
- (2) on the indifference curve, yielding the same utility level (when the IC is a straight line).



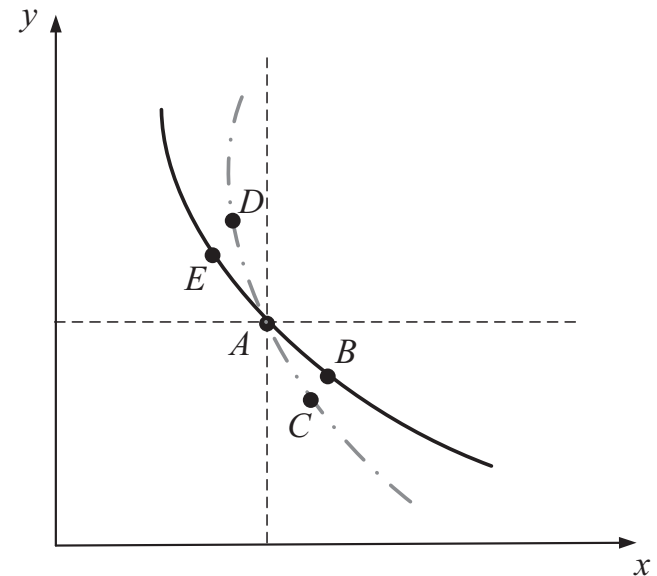
Properties of Indifference Curves

- **ICs cannot intersect.** It holds from monotonicity.
 - ICs in the figure intersect at bundle A , violating monotonicity.
- Bundle B lies northeast of C .
With monotonicity, $u_B > u_C$.
- Bundle D lies northeast of E .
With monotonicity $u_D > u_E$.
- Bundles C and D lie on the same IC, $u_C = u_D$. Similarly, $u_B = u_E$.



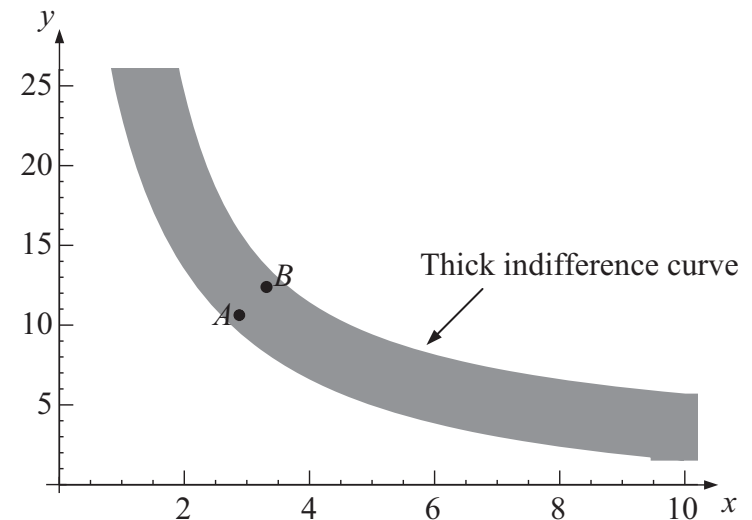
Properties of Indifference Curves

- ICs cannot intersect.
 - Combining with $u_B > u_C$,
$$u_E = u_B > u_C = u_D,$$
$$u_E > u_D,$$
which contradicts the result about bundles E and D ($u_D > u_E$).
 - Monotonicity $\rightarrow$ ICs cannot intersect.



Properties of Indifference Curves

- ICs are not thick. It holds from monotonicity.
 - The thick IC depicted in the figure violates monotonicity.
 - Bundles A and B lie in the same thick IC.
 - But, bundle B contains larger amounts of goods x and y than A . Then,
 - The consumer is not indifferent between A and B .
 - By monotonicity, $u_B > u_A$.
 - Monotonicity $\rightarrow$ ICs cannot be thick.



Marginal Rate of Substitution

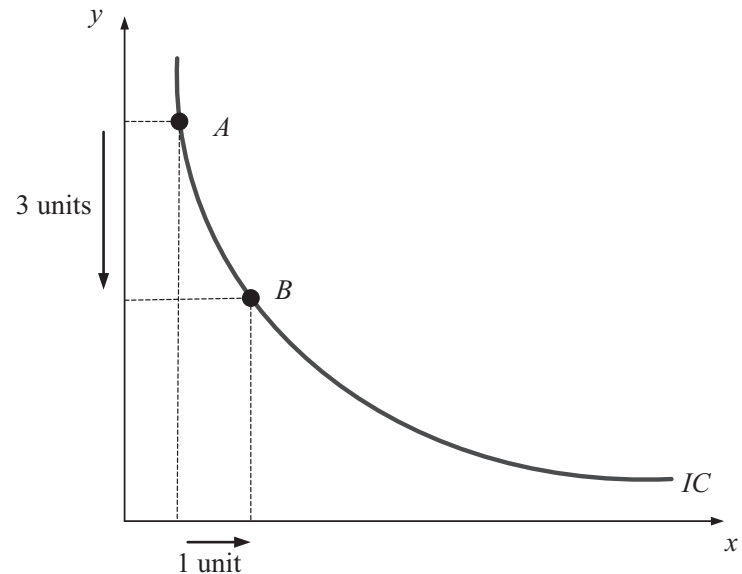
Marginal Rate of Substitution

- **Marginal rate of substitution (MRS)** is the rate at which a consumer is willing to give up units of good y as she receives an additional unit of good x , in order to keep utility level constant.

Formally,

$$MRS_{x,y} = \frac{MU_x}{MU_y}.$$

- MRS is the negative of the slope of the IC.
- Note that MRS is usually itself a function of x and y .

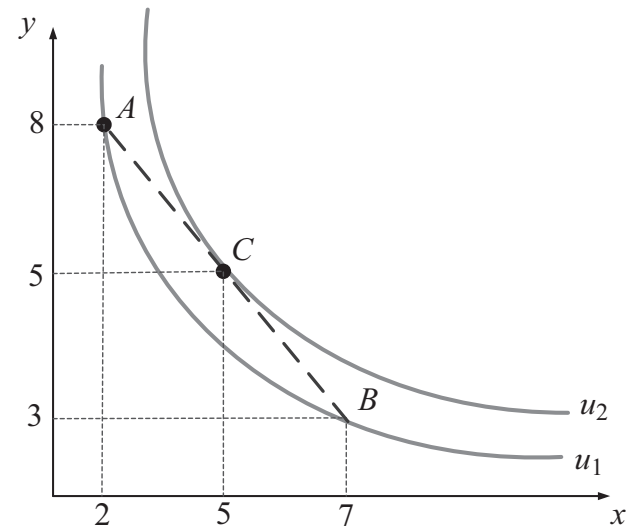


Marginal Rate of Substitution

- **Diminishing MRS.** The IC is relatively steep for small amounts of good x , but becomes flatter as we move rightward toward greater amounts of good x .

1. *Preference for variety.* Convex IC.

- The consumer is indifferent between extreme bundles, such as A and C , which yield utility level of u_1 .
- She prefers more balanced bundles, such as C , yielding a higher utility of u_2 .

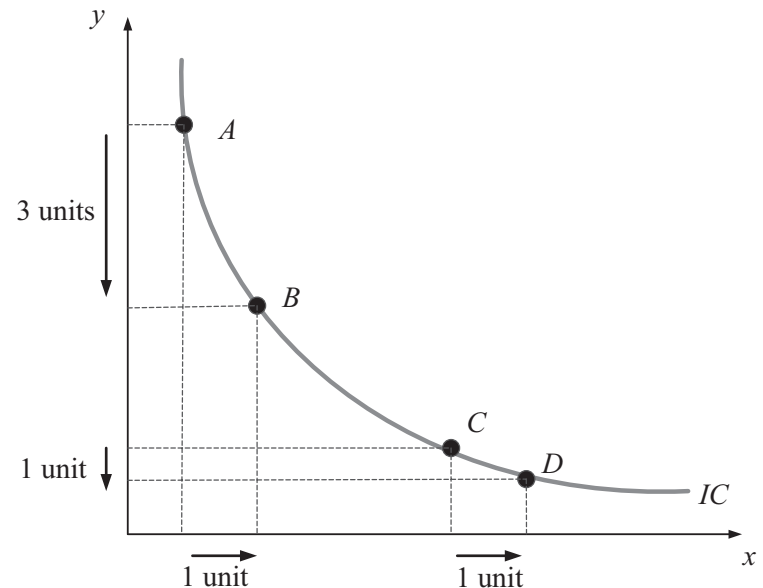


Marginal Rate of Substitution

- Diminishing MRS.

2. *Decreasing willingness to substitute.*

- At A , MU_x is high while MU_y is low.
 - The consumer is willing to give up several units of y to obtain more units of x .
- At C , MU_x is low and MU_y becomes high.
 - Willingness to give up units of y decreases once she has more units of x .



Marginal Rate of Substitution

- *Example: Finding MRS.*

1. Consider utility function $u(x, y) = x^{1/2}y^{1/2}$

$$MRS_{x,y} = \frac{MU_x}{MU_y} = \frac{\frac{1}{2} x^{-\frac{1}{2}} y^{\frac{1}{2}}}{\frac{1}{2} x^{\frac{1}{2}} y^{-\frac{1}{2}}} = \frac{y^{\frac{1}{2} - (-\frac{1}{2})}}{x^{\frac{1}{2} - (-\frac{1}{2})}} = \frac{y}{x}.$$

$MRS_{x,y}$ is decreasing in x , yielding ICs that are convex.

Marginal Rate of Substitution

- *Example* (continued):

2. Consider the linear utility function $u(x, y) = ax + by$ where $a, b > 0$,

$$MRS_{x,y} = \frac{MU_x}{MU_y} = \frac{a}{b}.$$

$MRS_{x,y}$ is constant in x .

Marginal Rate of Substitution

- *Example* (continued):

3. Consider utility function $u(x, y) = ax^2 + by^3$.

$$MRS_{x,y} = \frac{MU_x}{MU_y} = \frac{2ax}{3by^2}.$$

$MRS_{x,y}$ is increasing in x , yielding concave ICs. The IC is relatively flat for low values of x , but becomes steeper as we move rightward along the x -axis.

Special Types of Utility Functions

Special Types of Utility Functions

- Perfect Substitutes:

- Consider goods x and y . The consumer can use either good without significantly affecting her utility.

- *Examples:* Two brands of mineral water.

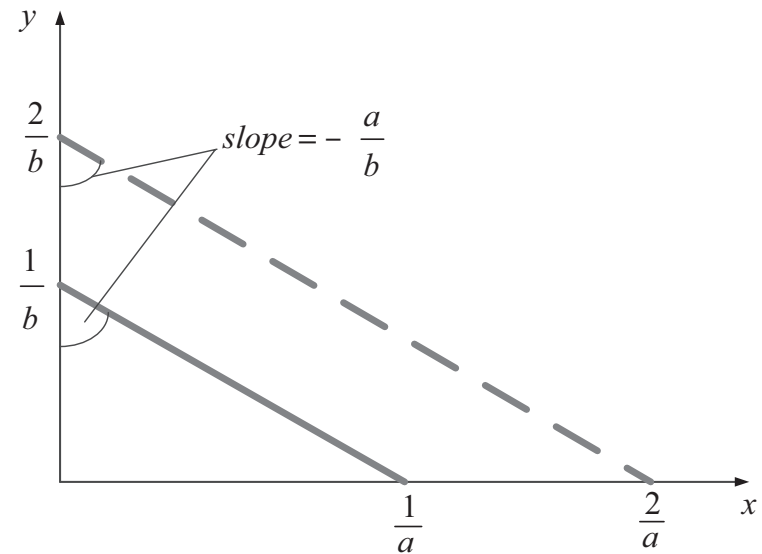
- The consumer's utility function takes the form

$$u(x, y) = ax + by, \text{ where } a, b > 0.$$

- This utility is linear in both goods.

Special Types of Utility Functions

- Perfect Substitutes (cont.):
 - ICs are straight lines:



Special Types of Utility Functions

- Perfect Substitutes (cont.):
 - Recall that MRS measures the consumer's willingness to give up units of good y to obtain 1 more unit of x , keeping her utility level unaffected.
 - A constant MRS (i.e., a number) $\rightarrow$ the consumer's willingness to substitute y for additional units of x is "always the same."
 - A decreasing MRS $\rightarrow$ the consumer is willing to give up more units of good y when x becomes relatively scarce.

Special Types of Utility Functions

- Perfect Complements:

- The consumer must consume goods in fixed proportions.
 - *Examples:* cars and petrol, left and right shoes.
- The utility function (referred as “Leontief”) takes the form

$$u(x, y) = A \min\{ax, by\}, \text{ where } A, a, b > 0.$$

- If $A = 1$ and $a = b = 2$, the utility function reduces to

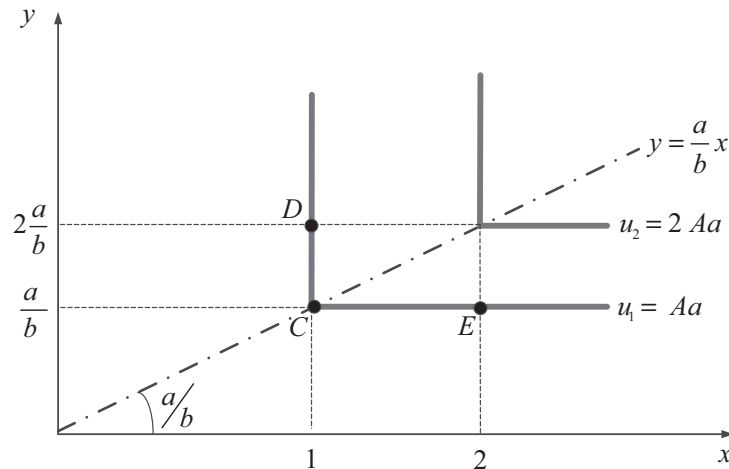
$$u(x, y) = \min\{2x, 2y\} = 2 \min\{x, y\}.$$

Special Types of Utility Functions

- **Perfect Complements** (cont.):
 - If the consumer increases the amount of x by 1 unit without increasing the amount of y , her utility does not necessarily increase.
 - If $x \geq y$, an increase in x does not increase her utility.
 - If $y > x$, an increase in x does increase her utility.

Special Types of Utility Functions

- **Perfect Complements** (cont.):
 - ICs have an L-shape:
 - The kink occurs at points where $ax = by$.
 - The slope is zero in the flat segment.
 - The slope is $-\infty$ in the vertical segment.



Special Types of Utility Functions

- Cobb-Douglas:

- The consumer regards goods x and y as neither perfectly substitutable nor complementary.
- The utility function takes the form

$$u(x, y) = Ax^\alpha y^\beta, \text{ where } A, \alpha, \beta > 0.$$

- Marginal utilities are

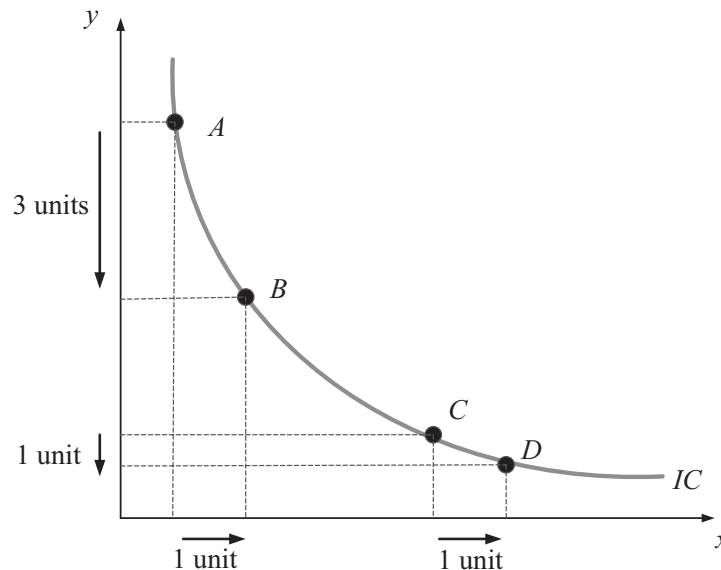
$$MU_x = A\alpha x^{\alpha-1}y^\beta \text{ and } MU_y = A\beta x^\alpha y^{\beta-1}.$$

which yield

$$MRS_{x,y} = \frac{MU_x}{MU_y} = \frac{A\alpha x^{\alpha-1}y^\beta}{A\beta x^\alpha y^{\beta-1}} = \frac{\alpha y^{\beta-(\beta-1)}}{\beta x^{\alpha-(\alpha-1)}} = \frac{\alpha y}{\beta x}.$$

Special Types of Utility Functions

- Cobb-Douglas (cont.):
 - $MRS_{x,y} = \frac{\alpha y}{\beta x}$ is decreasing in x .
 - ICs are convex, they become flatter as x increases.



Special Types of Utility Functions

- Cobb-Douglas (cont.):

- Special cases:

1. $A = \alpha = \beta = 1,$

$$u(x, y) = xy \Rightarrow MRS_{x,y} = \frac{y}{x}.$$

2. $A = 1, \alpha = \beta,$

$$u(x, y) = x^\alpha y^\alpha = (xy)^\alpha \Rightarrow MRS_{x,y} = \frac{y}{x}.$$

3. $A = 1, \beta = 1 - \alpha,$

$$u(x, y) = x^\alpha y^{1-\alpha} \Rightarrow MRS_{x,y} = \frac{\alpha}{1-\alpha} \frac{y}{x}.$$

Special Types of Utility Functions

- Quasilinear:

- Consumers who use all their additional income on one good alone, y (e.g., video games).
- Additional income is never spent on good x (e.g., toothpaste).
- This utility function takes the form

$$u(xy) = v(x) + by.$$

where $b > 0$, and $v(x)$ is a nonlinear function in x .

Special Types of Utility Functions

- Quasilinear (cont.):

- *Examples:*

- $v(x) = x^{1/2}$

- $v(x) = \ln x$.

- Any $v(x)$ which $v'(x)$ is not a constant, but instead depends on the units of good x , good y , or both.

- E.g., $v(x) = axy$, which $v'(x) = ay$ (not constant).

Special Types of Utility Functions

- Quasilinear (cont.):

- For $u(x, y) = v(x) + by$, the marginal utilities are $MU_x = v'(x)$ and $MU_y = b$, which yield

$$MRS_{x,y} = \frac{MU_x}{MU_y} = \frac{v'(x)}{b}.$$

- For a given value of x , the MRS is constant because it does not depend on the amount of good y .

Special Types of Utility Functions

- **Quasilinear** (cont.):
 - ICs are parallel shifts of each other.
 - If we fix constant the value of good x (e.g., $x = 16$), the slope of the IC ($MRS_{x,y}$) is unaffected by the amount of good y .

